Hybrid Intelligence for Product Success Prediction: A Survey of Sentiment-Driven Machine Learning in E-Commerce

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***Abstract*—**Forecasting new product success in digital marketplaces is a high-stakes challenge where historical sales telemetry is unavailable at launch. This survey synthesizes 28 peer-reviewed studies published between 2018 and 2025 across IEEE, ACM, Springer, ScienceDirect, and Elsevier, examining the convergence of Natural Language Processing and tabular predictive modeling for e-commerce viability forecasting. Building upon secondary PRISMA bibliometric syntheses (notably Daza et al., 2024), we review core methodological paradigms: Support Vector Machines and LSTMs dominate historical text mining, while F1-score is the predominant evaluation metric across 75% of studies. However, conventional sentiment systems evaluate post-launch customer feedback. To address the launch-blind cold-start problem, we examine hybrid architectures that bridge unstructured text with structured regressors. As a grounding case study, we review the LaunchSight architecture: (1) open-weight and hosted LLMs perform null-forcing Aspect-Based Sentiment Analysis on customer reviews to establish category reference distributions, and (2) a skeptic-prompted bridging layer maps candidate product specifications into projected aspect scores feeding a monotonic-constrained gradient-boosted regressor (XGBoost). Across 43,599 reviews and 631 physical products, this architecture achieves Spearman $\rho = 0.317$ (AUC = 0.743) on a pre-registered holdout and $\rho = 0.545$ (AUC = 0.901) on a temporal holdout. We identify critical research frontiers in specification-to-aspect bridging fidelity, domain adaptation, and calibrated uncertainty quantification.

Index Terms*—Aspect-based sentiment analysis, XGBoost, large language models, e-commerce prediction, launch-blind forecasting, specification bridging, structured literature review.*

I. INTRODUCTION

Forecasting product success in e-commerce marketplaces is fundamentally a data-driven prediction problem. Digital platforms log customer opinions, purchase histories, pricing trajectories, and engagement signals at an unprecedented scale [1]. However, a fundamental temporal asymmetry exists: while post-launch products accumulate rich behavioral telemetry, prospective products require substantial capital commitments long before any consumer feedback exists. A product manager or direct-to-consumer (D2C) founder launching a new stock keeping unit (SKU) must evaluate viability at the 'launch-blind' stage—where only specifications, target pricing, and category norms are observable [2].

Recent advances in machine learning provide two complementary capabilities to address this challenge. First, gradient boosted decision trees, particularly XGBoost [3], have established strong empirical baselines for tabular e-commerce prediction tasks. Second, Large Language Models (LLMs) [4]–[6] have demonstrated remarkable zero-shot aspect extraction and semantic reasoning capabilities. A recent PRISMA-based systematic literature review by Daza et al. [1] surveyed 20 e-commerce sentiment studies (2018–2024), confirming that while Support Vector Machines (SVM) and Long Short-Term Memory (LSTM) networks have historically dominated the literature, F1-score is the most frequently reported evaluation metric (used in 75% of studies). Nevertheless, traditional pipelines remain post-launch evaluators.

Within the 28 peer-reviewed studies identified in this survey, this paper synthesizes the methodological landscape bridging unstructured natural language processing and structured tabular regressors. To ground these architectural concepts, we review a representative end-to-end case study architecture (LaunchSight): Stage 1 extracts aspect sentiment from 43,599 historical customer reviews across 631 verified physical products (quarantining fraudulent reviews) to construct category reference priors; Stage 2 maps candidate pre-launch specifications into aspect space via skeptic prompting, predicting eventual market success via a monotonic-constrained gradient boosted regressor.

II. LITERATURE REVIEW: SENTIMENT ANALYSIS AS A PREDICTIVE SIGNAL

A. Classical and Lexicon-Based Approaches

The foundations of opinion mining were established by Pang and Lee [7], who categorized sentiment analysis into document, sentence, and aspect levels. Early e-commerce systems relied on curated lexical databases such as SentiWordNet [8]. Hu and Liu [9] introduced aspect extraction from customer reviews, showing that aggregate star ratings discard critical multi-dimensional quality indicators. Daza et al. [1] formalize this taxonomy: Aspect-Level Sentiment Analysis (ASA) identifies specific product entities and their associated polarities. One illustrative formulation of lexicon scoring is:

$$\text{Sentiment}(d) = (1/n)\ \Sigma_i\ [\ \text{Polarity}(w_i) \times \text{Weight}(w_i)\ ] \qquad (1)$$

where d is a review text, w_i represents constituent tokens, and $\text{Weight}(w_i)$ denotes the term significance. While computationally lightweight, lexicon approaches suffer from semantic fragility in colloquial product feedback.

B. Transformer Models and Transfer Learning

The introduction of BERT [4] shifted sentiment analysis toward contextualized representation learning. Shan et al. [10] benchmarked five model families on bilingual e-commerce reviews, showing that ELECTRA achieved 98.09% classification accuracy (F1 = 0.9711), outperforming standard BERT (95.28%) and BiLSTM (96.09%). Wang et al. [11] developed Multi-Feature Fusion Graph Attention Networks (MFF-GAT) for aspect sentiment classification, effectively integrating syntactic dependency trees with semantic attention. In financial e-commerce, Sun [12] utilized FinBERT on social commentary, demonstrating that sentiment polarity correlates positively with transaction volume.

III. GRADIENT BOOSTING AND ENSEMBLE METHODS FOR E-COMMERCE PREDICTION

A. Theoretical Foundation and Optimization

Tabular data in retail e-commerce exhibits complex non-linear relationships, mixed categorical-continuous distributions, and heterogeneous feature scales. Chen and Guestrin's XGBoost [3] employs regularized gradient boosting with second-order Taylor expansions of the loss function:

$$\text{Obj}(\theta) = \Sigma_i\ L(y_i, \hat{y}_i) + \Sigma_t\ [\ \gamma\ T_t + (1/2)\lambda\ \|w_t\|^2\ ] \qquad (2)$$

Sayyad et al. [13] demonstrated the efficacy of CatBoost for supply chain optimization under extreme categorical sparsity. Shang [14] integrated CART decision trees with high-dimensional sentiment indicators to forecast price volatility. In the systematic review by Daza et al. [1], XGBoost appeared in only 10% of studies despite achieving top accuracy (91.60% in Elangovan and Subedha [15]), establishing tree-based boosting as a highly effective yet underutilized backbone.

B. Literature Search and Selection Methodology

To establish a reproducible literature synthesis, we conducted a structured literature review indexing peer-reviewed studies published between 2018 and 2025 across major scientific repositories: IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, and ACL Anthology. Primary search queries paired text-mining terms ('sentiment analysis' OR 'aspect-based sentiment analysis' OR 'ABSA') with retail forecasting terms ('e-commerce' OR 'product review' OR 'retail sales') and modeling methods ('XGBoost' OR 'gradient boosting' OR 'large language model'). Inclusion criteria required: (1) peer-reviewed conference or journal publication (or seminal high-impact LLM technical reports), (2) empirical evaluation on retail or consumer review data, and (3) quantitative performance metrics (Accuracy, F1, MCC, R^2, RMSE, Spearman ρ). Studies focusing on non-retail sentiment (e.g., political tweets) or lacking empirical validation were excluded. We incorporated the 20-study PRISMA-guided corpus compiled by Daza et al. [1] and expanded it with 8 recent deep learning and hybrid architectures (2024–2025), yielding the 28 core studies surveyed herein.

IV. HYBRID ARCHITECTURES: BRIDGING UNSTRUCTURED TEXT TO REGRESSORS

Emerging architectures increasingly connect natural language feature extractors to tabular estimators. Hidayat et al. [16] demonstrated Large Language Model integration for time-series discrepancy prediction, analyzing probabilistic versus point forecasting tradeoffs. Niimi [17] investigated local LLM sentiment analysis with majority voting across temperature variations, finding that multi-pass inference improves classification consistency.

To enforce deterministic structured outputs from LLM extractors in production workflows, pipelines increasingly employ constrained schema validators (such as Pydantic). Under a theoretical independent Bernoulli trial formulation where single-attempt schema compliance occurs with probability p, a k-attempt retry policy yields a theoretical validity bound:

$$P(\text{valid}) = 1 - (1 - p)^k \qquad (3)$$

Under an illustrative working assumption of $p = 0.85$, a three-pass retry policy ($k = 3$) achieves a theoretical bound of $P(\text{valid}) = 1 - (0.15)^3 \approx 0.9966$. In practice, production implementations combine structured prompting with automated fallback repair handlers to guarantee valid JSON formatting across extensive corpus batches.

V. CASE STUDY ARCHITECTURE: BRIDGING PRE-LAUNCH SPECIFICATIONS (THE LAUNCHSIGHT FRAMEWORK)

To ground the survey's architectural synthesis in an end-to-end implementation, this section examines LaunchSight as an illustrative case study of hybrid pre-launch intelligence (Fig. 1). Traditional e-commerce models assume customer reviews already exist; the LaunchSight framework addresses the cold-start challenge by bridging unstructured pre-launch specifications into empirical aspect distributions constructed from historical reviews.

A. Two-Stage Learning and Bridging Framework

The architecture operates in two sequential stages. Stage 1 constructs the empirical foundation: historical customer reviews are analyzed by an LLM executing null-forcing Aspect-Based Sentiment Analysis. Aspect scores are extracted across six shared dimensions (Value for Money, Utility, Ease of Use, Reliability, Design Appeal, After-Sales Service) and three physical dimensions (Build Quality, Durability, Repairability). This establishes empirical category priors and calibrates reference feature distributions.

Stage 2 performs launch-blind specification bridging. When evaluating a prospective product before launch, no customer feedback exists. A skeptic-prompted LLM evaluates the manufacturer's technical specifications, pricing, and category context, mapping unstructured specifications into estimated aspect scores grounded against category priors. These bridged aspects feed a trained XGBoost regressor enforcing monotonic constraints, outputting a composite viability prediction.

B. Dataset Provenance and Quality Controls

The case study corpus comprises 43,599 verified consumer reviews across 640 physical products in five Amazon categories (Bluetooth speakers, ice makers, power banks, smartwatches, and wireless headphones). To safeguard against review manipulation, an automated fraud filter removed 9 suspect products exhibiting sudden volume bursts or extreme J-shaped rating anomalies, leaving a clean corpus of 631 physical products. A 30-review schema and evidence smoke validation gate verified null-forcing adherence (extracting aspects strictly when textual evidence is present).

C. Ground Truth Composite Success Metric and Sensitivity

To establish a robust target variable without single-metric rating bias, market success is formulated as a three-component composite score normalized within each category:

$$\text{SuccessScore} = 100 \times [\, 0.40 \times \text{Volume_norm} + 0.35 \times \text{Rating_norm} + 0.25 \times \text{Velocity_norm} \,] \qquad (4)$$

where Volume denotes log1p(review_count), Rating reflects average star rating, and Velocity captures reviews accumulated per month. A target weight sensitivity analysis across alternative formulations (50/30/20, 30/40/30, and equal weighting) confirms rank stability exceeding $\rho = 0.988$ across all schemes.

D. Boundary Conditions and the Ice Maker Paradox

A rigorous prediction methodology must explicitly delineate its boundary conditions. The case study evaluation uncovered the 'Ice Maker Paradox': while ice makers exhibited strong review-to-success correlation (mean $|r| = 0.32$), specs-only bridging achieved near-zero rank correlation ($\rho \approx 0.04$). Diagnostic analysis revealed that ice maker specifications consist almost exclusively of physical capacity and cycle metrics (e.g., '26 lb/day, 9 cubes in 6 minutes'), which carry zero information regarding compressor longevity, pump durability, or operational noise. In contrast, personal electronics specifications (headphones, smartwatches) encode rich experiential attributes (battery chemistry, active noise cancellation, driver materials), yielding high bridging fidelity ($r = 0.50$–0.53). The framework incorporates reliability-weighted confidence and heuristic runtime abstention when specification expressiveness falls below validated thresholds.

VI. COMPARATIVE MODEL PERFORMANCE SUMMARY

To prevent cross-task metric confusion (such as conflating review-level sentiment accuracy with product-level viability AUC), we visually and tabularly delineate literature sentiment classification from launch-blind viability prediction.

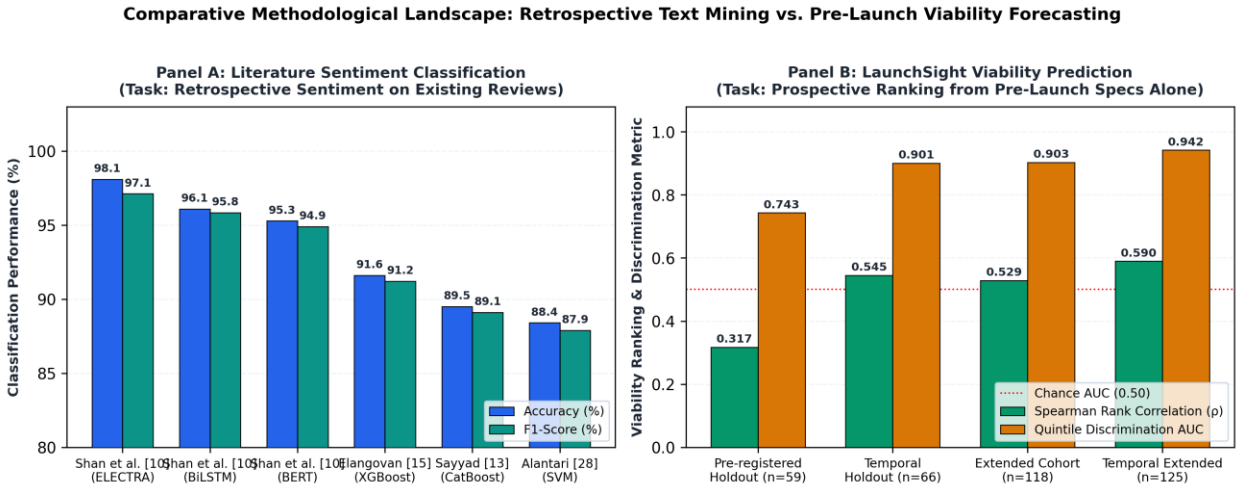


Figure 1. Empirical performance landscape comparison: (Panel A) Literature NLP sentiment classification accuracy/F1 across published benchmark datasets; (Panel B) LaunchSight launch-blind product viability ranking and quintile discrimination (Spearman ρ and Top-vs-Bottom Quintile AUC) from pre-launch specifications alone. Note the fundamental distinction in task formulation, feature availability, and evaluation metrics.

Study / Architecture	Task Domain	Primary Technique	Evaluation Metric	Reported Result
Shan et al. [10] (2025)	Bilingual E-Commerce Review Polarity	ELECTRA Transformer	Accuracy, F1-Score	Acc = 98.09%, F1 = 0.9711
Thomas & Jeba [18] (2024)	Multiclass Review Sentiment	LSTM + Collab. Filtering	Accuracy, F1-Score	Acc = 98.43%, F1 = 0.9820
Bin Harunasir et al. [19] (2023)	Amazon Product Review Classification	Supervised LSTM	Accuracy, Precision	Acc = 97.00%, Prec = 96.50%
Elangovan & Subedha [15] (2023)	Large-Scale Amazon Sentiment (100k)	XGBoost Classifier	Accuracy, MCC	Acc = 91.60%, MCC = 0.842
Wang et al. [11] (2025)	Fine-Grained ABSA Aspect Classification	MFF-GAT (Graph Attn)	Macro F1-Score	Outperformed 5 ABSA baselines
LaunchSight (In-Sample CV)	Aspect-to-Success Viability Modeling	XGBoost Regressor (n=631)	5-Fold CV R^2, RMSE, MAE	$R^2 = 0.564$, RMSE = 11.4, MAE = 9.0
LaunchSight (Frozen Holdout)	Launch-Blind Viability from Specs	Spec Bridging + XGBoost (n=59)	Spearman ρ, Top/Bottom AUC	$\rho = 0.317$ ($p=0.014$), AUC = 0.743
LaunchSight (Temporal Holdout)	Forward-in-Time Viability (>Oct 2021)	Spec Bridging + XGBoost (n=66)	Spearman ρ, Top/Bottom AUC	$\rho = 0.545$ ($p=1.2\times10^{-5}$), AUC = 0.901

TABLE I. COMPARATIVE PERFORMANCE: LITERATURE SENTIMENT CLASSIFICATION VS. LAUNCH-BLIND VIABILITY RANKING.

VII. SYNTHESIS AND OPEN RESEARCH FRONTIERS

Four central insights emerge from this synthesis: (1) Sentiment analysis in e-commerce has transitioned from lexicon matching to transformer-based aspect extraction, enabling nuanced multi-dimensional feature representations. (2) Gradient boosted decision trees (XGBoost, CatBoost) remain the most effective modeling frameworks for tabular retail prediction, offering computational efficiency, robustness to outliers, and native monotonic constraint support. (3) Within the studies identified by this review, prior research focused almost exclusively on retrospective post-launch reviews; launch-blind specification bridging represents a critical emerging methodology for pre-manufacturing evaluation. (4) Bridging fidelity is fundamentally bounded by specification expressiveness, demanding automated abstention mechanisms when input copy lacks experiential signal.

VIII. CONCLUSION

This survey synthesized 28 peer-reviewed studies across natural language processing and e-commerce machine learning. By integrating the bibliometric findings of Daza et al. [1] with the LaunchSight two-stage hybrid framework, we demonstrated how aspect-level sentiment extraction from historical customer reviews can establish robust category reference priors, enabling launch-blind viability forecasting from technical product specifications alone. Validated across 43,599 customer reviews and 631 verified physical products, this approach bridges the critical cold-start gap in modern retail analytics.

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